Aim:- Write a java program to implement interface for achieving abstraction & multiple inheritence through programs.

Theory:- An ~~inherit~~ interface in Java is a blueprint of a class that contains abstract methods (methods) without body) & constants. It is used to achieve abstraction & supports multiple inheritence in Java

An interface is declared using the interface keyword, and all its methods are public & abstract by default.

Syntax

```
interface InterfaceName{
    void method1();
    void method2(); }
```

Implementing an Interface → A class implements an interface using the implements keyword & must provide definitions for all its methods.

Features - Support ~~interface~~ abstraction.
- Allows multiple inheritence
- Methods are public & abstract by default
- Variables are public, static & final.

Types of interface -

- Normal interface
- Functional interface (contains one abstract method)
- Marker Interface (empty interface)

Program 1. To demonstrate how to implement an interface in Java.

```
interface Animal {
        void sound(); //abstract method
    }
    class Dog implements Animal {
            public void sound() {
            System.out.println("Dog barks");
        }
    }

    public class InterfaceExample {
        public static void main (String [] args) {
            Dog d = new Dog ();
            d.sound();
        }
    }
```

Program 2 : To implement multiple methods from an interface -

```
I. interfare Shape {
        void area();
        void perimeter();
    }
```

Output

Area = 50
perimeter = 30


```
class Rectangle implements shape {
    int length =10;
    int wide = 5;

    public void area() {
        System.out.println("Area = " + (length * width));
    }
    public void perimeter() {
        System.out.println("perimeter = " + 2*(length + width));
    }
}
public class InterfaceMultiple {
    public static void main(String[] args) {
        Rectangle r = new Rectangle();
        r.area();
        r.perimeter();
    }
}
```

Program 3 To demonstrate multiple ~~interfacein~~ inheritance using interfaces.

```
interface A {
    void show();
}
interface B {
    void Display();
}
class Test implements A, B {
    public void show() {
```

output

Method from interface A
Method from interface B.

Expt. No.

Date


```
System.out.println("Method from interface A");
}
public ~~static~~ void Display(){
        System.out.println("Method from interfaceB");
    }
}

public class MultipleInterfaceExample{
    public static void main(String[] args){
    Test t = new Test();
      t.show();
      t.Display();
    }
}
```

Viva Questions -

Q1 What is an interface in Java?
Ans An interface in Java is a reference type that contains abstract methods & constants. It is used to achieve abstraction & supports multiple inheritance in Java program.

Q2 How does an interface achieve abstraction?
Ans An interface acheives abstraction by declaring abstraction methods without implementation. The implementation is provided by the class that implements the interface, hiding internal details from the user.

Teacher's Signature

Q3 Why does Java use Interfaces for multiple inheritance?
Ans Java uses interfaces for multiple inheritance because classes cannot extend multiple classes directly. Interfaces avoid ambiguity problems & allow a class to inherit features for multiple sources.

Q4. What ~~do~~ is the difference between a class & an interface?
Ans A class can contain both implemented & abstracts methods, while an interface mainly contains abstracts method. A class is extended, whereas an interface is implemented using the implements keyword.

Q5. Can a class implement more than one interface?
Ans Yes, a class can implement multiple interfaces in Java. This helps achieve multiple inheritance & allows one class to provide implementations for methods from different interfaces.

Aim :- To study and implement graphics programming using applet in Java for drawing shapes, colors, & simple graphical elements.

Theory :- Java provides the applet class to create graphical applications that can run ~~outside~~ inside a web browser or applet viewer. Graphics programming in Java is done using the java.awt package, which provides classes for drawing shapes, colors & text.

An applets is a small Java program that runs inside a browser or applet viewer. It does not have a main() method & is controlled by the browser.

Life Cycle of Applet :- An applet goes through different stages during execution :-

- init() → initializes the applet
- start() → starts execution
- paint(Graphics g) → used for drawing graphics
- Stop() → stops execution
- Destroy() → Cleans up resources.

Graphics in Java :- Graphics in Java are handled using the Graphics class from the Java.awt package.

- drawLine() → Draws a line
- drawRect() → Draws a rectangle

- drawOval() → draws an oval
- drawString() → Displays text
- setColor() → Sets drawing color

Features of Applet

- Supports graphics & animation
- Platform independent
- Event-driven execution

P1 To demonstrate drawing shapes using Applet & graphics class -

```
import java.applet.Applet;
import java.awt.Graphics;
import java.awt.Color;

/*
 <applet code = "GraphicsApplet" width = 400
       Height = 300>
  </applet>
 */
 public class GraphicsApplet extends Applet {
   public void paint (Graphics g){

//Set Color
  g.setColor (Color.RED);
  g.drawString ("Hello Applet Graphics", 50, 50);
// Drawline
 g.drawLine (50, 70, 200, 70);
```

Teacher's Signature

output:

~~st 1~~

step 1: compile

javac GraphicsApplet.java

Step 2: Run using Applet Viewer

appletviewer GraphicsApplet.java


```
//Draw rectangle
  g.setColor(Color.BLUE);
  g.drawRect(50,100,100,50);
//Draw filled rectangle
  g.setColor(Color.GREEN);
  g.fillRect(200,100,100,50);

//Draw Oval
  g.setColor(Color.MAGENTA);
  g.drawOval(50,180,100,50);

//Draw filled oval
  g.setColor(Color.~~ORANGE~~ ORANGE);
  g.fillOval(200,180,100,50);
  }
}
```

Viva Question :-

Q1 What is an applet in Java?
Ans An applet is a small Java program that runs inside a web browser or applet viewer to create graphical & interactive application.

Q2 Which package is used for graphics in Java applets?
Ans The java.awt package is used for graphics in Java applets. It provides classes for drawing shapes, colors & fonts.

Q3 Which methods is used for drawing in an applets?
Ans The paint(Graphics g) method is used for drawing graphics such as lines, rectangles, circles & text in an applets.

Q4 What is the use of the Graphic class?
Ans The graphics class provide methods to draw shapes, text & images inside an applet or graphical Java application.

Q5 What are some common drawing methods in Graphics class?
Common drawing methods are:
- drawLine()
- drawRect()
- drawOval()
- drawString()
- fillRect()